

Companion to Bird & de Moor, Algebra of Programming

§8. Thinning Algorithms

§8.1. Thinning

(8.1)

For $Q : A \rightarrow A$, $\text{thin}(Q) \triangleq (\exists/\exists) \cap (\epsilon \setminus (Q^\circ \epsilon)) : EA \rightarrow EA$ (8.1).

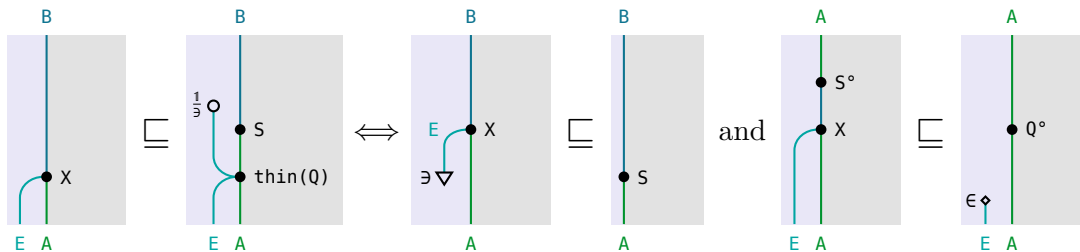
$ys \text{ thin}(Q) xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \ Q \ a)$ the same $^\circ$ as $\triangleq \text{est}$: ϵX runs $a \rightarrow ys \rightarrow xs$, member of ys first, and $Q^\circ \epsilon$ runs $a \rightarrow b \rightarrow xs$, so $(a, b) \in Q^\circ$ reads $b \ Q \ a$; at $Q^\circ \leq$ every $a \in ys$ keeps some $b \leq a$ in xs , the end $\text{est}(\leq)$ picks

(8.1a)

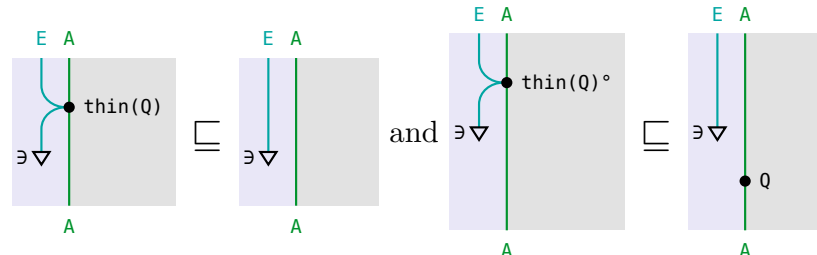
the law	what it says
$X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \exists \exists S$ and $S^\circ X \sqsubseteq Q^\circ \epsilon$	everything kept is an S-value, and every S-value has a Q-lower bound among the kept ones
$\text{thin}(Q) \exists \sqsubseteq \exists$	everything a thinning keeps was in the set
$\text{thin}(Q)^\circ \exists \sqsubseteq \exists Q$	every element of the set has a Q-lower bound among the kept ones
$Q \sqsubseteq R \implies \text{thin}(Q) \sqsubseteq \text{thin}(R)$	the fewer pairs Q relates, the fewer subsets count as thinnings
$1 \sqsubseteq Q \implies 1 \sqsubseteq \text{thin}(Q)$ $Q \ Q \sqsubseteq Q \implies \text{thin}(Q) \ \text{thin}(Q) \sqsubseteq \text{thin}(Q)$	keeping everything is always a legal thinning
$Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge R \ R \sqsubseteq R \implies \text{est}(R) = \text{thin}(Q) \ \text{est}(R)$ $Q \sqsubseteq R, 1 \sqsubseteq Q, R R \sqsubseteq R \implies$ (8.1.2a); weaker than the book's "both preorders": Q transitive is never used and $1 \sqsubseteq R$ follows	thin-introduction: thinning first cannot lose an R-minimum
$\text{thin}(Q) \sqsupseteq \text{est}(Q) \ \frac{1}{\exists}$ (8.2) — (8.2)	thin-elimination: keeping one element is a thinning, but its domain is the sets $\text{est}(Q)$ is defined on
$\frac{S}{\exists} \text{thin}(Q) \sqsupseteq \frac{S}{\exists} \text{est}(R) \ \frac{1}{\exists}$ (8.3), $R \cap (S^\circ S) \sqsubseteq Q$ — thin-elimination variant	the usable variant: R need only refine Q between values S gives one argument
$P(\text{thin}(Q)) \text{ union} \sqsubseteq \text{union thin}(Q)$ (8.4)	thinning each member set is a thinning of the union

§8.1.1. $X \sqsubseteq (S^\circ \exists) \text{thin}(Q) \iff X \exists \exists S$ and $S^\circ X \sqsubseteq Q^\circ \epsilon$

(8.1.1a)



(8.1.1b)



(8.1.1a) at $X \triangleq \text{thin}(Q)$, $S \triangleq \exists$: the left side is $\text{thin}(Q) \sqsubseteq \text{thin}(Q)$, since $\exists \exists \exists = \Lambda(\exists) = 1$

§8.1.2. $\text{est}(R) = \text{thin}(Q) \ \text{est}(R)$ given $Q \sqsubseteq R, 1 \sqsubseteq Q, R R \sqsubseteq R$

$\text{est}(R) \sqsubseteq \text{thin}(Q) \text{ est}(R)$ (8.1.2a)
the $\sqsubseteq$ half: keeping everything is a thinning — $1 \sqsubseteq Q$

(a) (b)

(a)	$\text{est}(R)$	$1 \sqsubseteq \text{thin}(Q)$ — (8.1a)
(b)	$\text{thin}(Q) \text{ est}(R)$	

$\text{thin}(Q) \text{ est}(R) \sqsubseteq \exists$ (8.1.2b)
the $\sqsubseteq$ half, first condition of the UP of est at $X \triangleq \text{thin}(Q) \text{ est}(R)$ — est-up

(a) (b) (c)

(a)	$\text{thin}(Q) \text{ est}(R)$	$\text{est}(R) \sqsubseteq \exists$ — est-laws
(b)	$\text{thin}(Q) \exists$	$\text{thin}(Q) \exists \sqsubseteq \exists$ — (8.1.1a)
(c)	$\exists$	

$\epsilon \text{ thin}(Q) \text{ est}(R) \sqsubseteq R^\circ$ (8.1.2c)
the $\sqsubseteq$ half, second condition — $Q \sqsubseteq R$, R transitive

(a) (b) (c) (d) (e)

(a)	$\epsilon \text{ thin}(Q) \text{ est}(R)$	$\epsilon \text{ thin}(Q) \sqsubseteq Q^\circ \epsilon$ — (8.1.1a)
(b)	$Q^\circ \epsilon \text{ est}(R)$	$\epsilon \text{ est}(R) \sqsubseteq R^\circ$ — est-laws
(c)	$Q^\circ R^\circ$	$Q \sqsubseteq R$
(d)	$R^\circ R^\circ$	R transitive
(e)	R°	

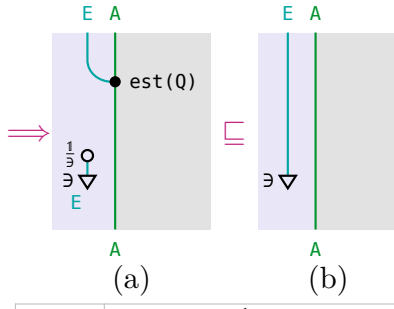
$$\text{est}(Q) \stackrel{1}{\exists} \sqsubseteq \text{thin}(Q)$$

(8.2)

the singleton holding a Q -least member of a set is a thinning of that set

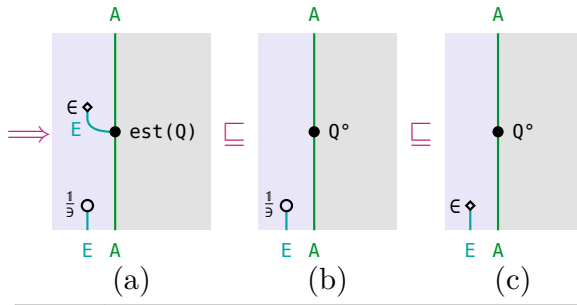
$$\text{est}(Q) \stackrel{1}{\exists} \ni \sqsubseteq \ni$$

every member of the singleton is a member of the set



(a)	$\text{est}(Q) \stackrel{1}{\exists} \ni$	$\stackrel{1}{\exists} \ni = 1$ — pow-laws; $\text{est}(Q) \sqsubseteq \ni$ — est-laws
(b)	$\ni$	$\triangleq \text{thin}, \epsilon \setminus (Q^\circ \epsilon)$ half

$$\epsilon \text{ est}(Q) \stackrel{1}{\exists} \sqsubseteq Q^\circ \epsilon$$

the singleton's member stands in Q to every member of the set

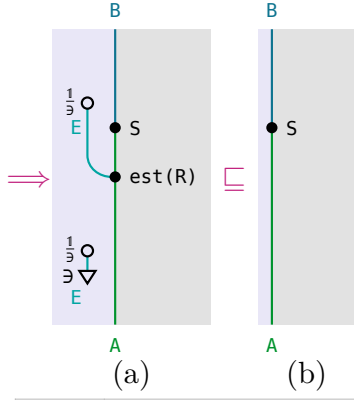
(a)	$\epsilon \text{ est}(Q) \stackrel{1}{\exists}$	$\epsilon \text{ est}(Q) \sqsubseteq Q^\circ$ — est-up
(b)	$Q^\circ \stackrel{1}{\exists}$	$\stackrel{1}{\exists} \sqsubseteq \epsilon$ — pow-laws
(c)	$Q^\circ \epsilon$	

$$R \cap S^\circ S \sqsubseteq Q \Rightarrow \frac{S}{S} \text{est}(R) \frac{1}{S} \sqsubseteq \frac{S}{S} \text{thin}(Q) \quad (8.3)$$

given $Rn(S^\circ S) \sqsubseteq Q$, Q a preorder

$$\frac{S}{S} \text{est}(R) \frac{1}{S} \ni \sqsubseteq S$$

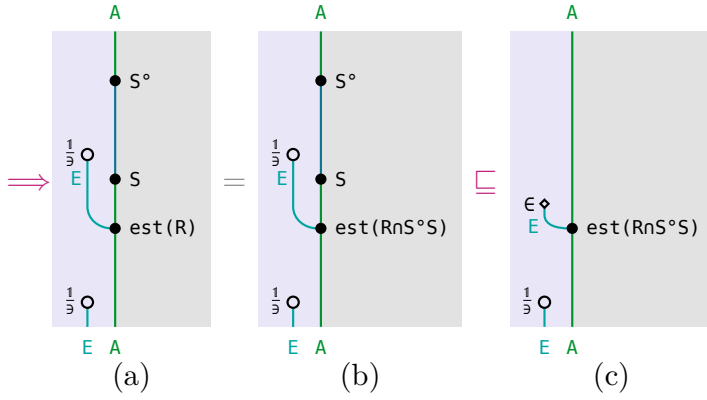
every member of the singleton is a value S returns



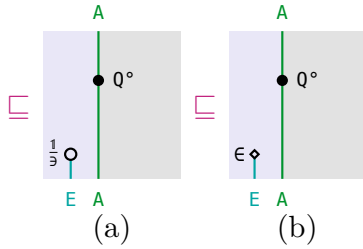
(a)	$\frac{S}{S} \text{est}(R) \frac{1}{S} \ni$	$\frac{1}{S} \ni 1$ — pow-laws; $\frac{S}{S} \text{est}(R) \sqsubseteq S$ — est-laws
(b)	S	(8.1a), second condition

$$R \cap S^\circ S \sqsubseteq Q \Rightarrow S^\circ \frac{S}{S} \text{est}(R) \frac{1}{S} \sqsubseteq Q^\circ \in$$

the singleton's member stands in Q to every value S returns at the same argument



(a)	$S^\circ \frac{S}{S} \text{est}(R) \frac{1}{S}$	$\frac{S}{S} \text{est}(R) = \frac{S}{S} \text{est}(RnS^\circ S)$ — est-laws
(b)	$S^\circ \frac{S}{S} \text{est}(R \cap S^\circ S) \frac{1}{S}$	$S^\circ \frac{S}{S} \sqsubseteq E$ — pow-laws
(c)	$E \text{est}(R \cap S^\circ S) \frac{1}{S}$	$E \text{est}(RnS^\circ S) \sqsubseteq (RnS^\circ S)^\circ$ — est-up; $Rn(S^\circ S) \sqsubseteq Q$



(a)	$Q^\circ \frac{1}{S}$	$\frac{1}{S} \sqsubseteq E$ — pow-laws
(b)	$Q^\circ E$	

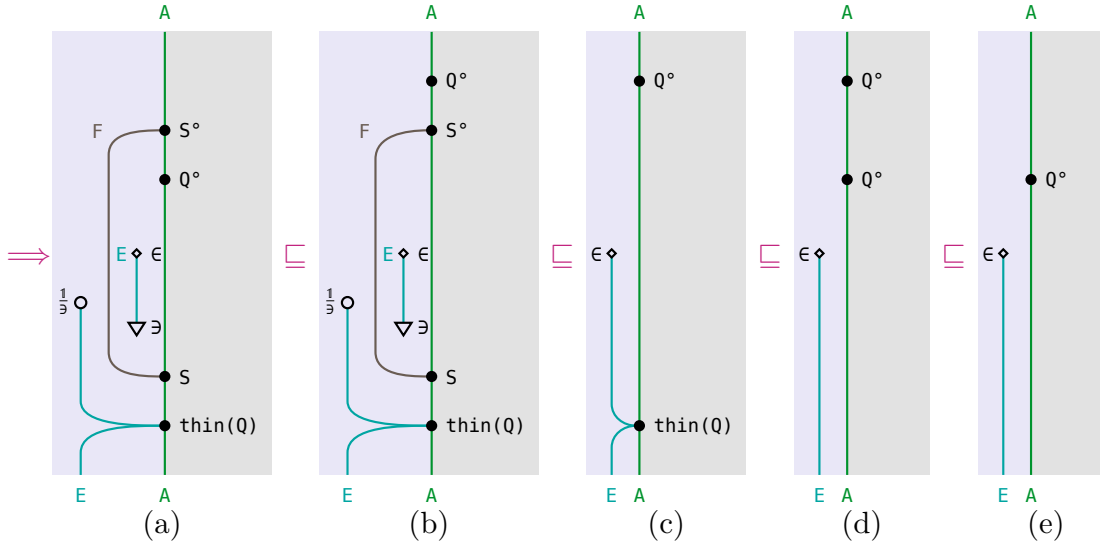
$$Q \subseteq Q \wedge \text{MonoAlg}(S, Q) \Rightarrow (\frac{F(\exists) S}{\exists} \text{thin}(Q)) \subseteq \frac{(S)}{\exists} \text{thin}(Q)$$

Theorem 8.1

thinning at every step of the reduce is a thinning of the whole candidate set — S monotonic on Q , Q a preorder

$$(\frac{F(\exists) S}{\exists} \text{thin}(Q)) \supseteq (S) \text{ and } (S) \circ (\frac{F(\exists) S}{\exists} \text{thin}(Q)) \subseteq Q^\circ \in$$

(8.1a) at $X \triangleq (\frac{F(\exists) S}{\exists} \text{thin}(Q))$, (S) for its S

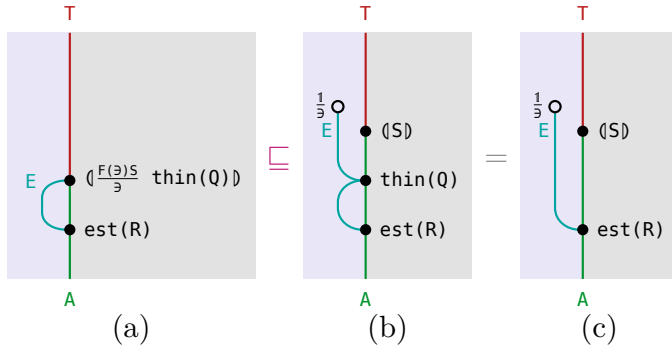


(a)	$S^\circ F(Q^\circ \in) \frac{F(\exists) S}{\exists} \text{thin}(Q)$	$S^\circ F(Q^\circ) \subseteq Q^\circ S^\circ$ — mon-str, relator-laws
(b)	$Q^\circ S^\circ F(\in) \frac{F(\exists) S}{\exists} \text{thin}(Q)$	$S^\circ F(\in) \frac{F(\exists) S}{\exists} \subseteq \in$ — pow-laws
(c)	$Q^\circ \in \text{thin}(Q)$	$\in \text{thin}(Q) \subseteq Q^\circ \in$ — $\triangleq$ thin, adj-all
(d)	$Q^\circ Q^\circ \in$	Q a preorder
(e)	$Q^\circ \in$	

$$Q \subseteq R \wedge 1 \subseteq Q \wedge Q \subseteq Q \wedge R \subseteq R \wedge \text{MonoAlg}(S, Q) \Rightarrow (\frac{F(\exists) S}{\exists} \text{thin}(Q)) \text{ est}(R) \subseteq \frac{(S)}{\exists} \text{est}(R)$$

Corollary 8.1

the thinning fold refines the optimisation problem itself — S monotonic on Q , $Q \subseteq R$, both preorders



(a)	$(\frac{F(\exists) S}{\exists} \text{thin}(Q)) \text{ est}(R)$	thinning theorem
(b)	$\frac{(S)}{\exists} \text{thin}(Q) \text{ est}(R)$	$\text{est}(R) = \text{thin}(Q) \text{ est}(R)$ — (8.1a)
(c)	$\frac{(S)}{\exists} \text{est}(R)$	

§8.2. Paths in a layered network

(8.2a)

$F(A, X) = A + A \times X$, $L = \text{list}^+$ with initial algebra $\alpha \triangleq [\text{wrap}, \text{cons}] : F(A, LA) \rightarrow LA$.

$\text{wrapz} \triangleq \langle \text{wrap}, \text{zero} \rangle$, $\text{consw}(a, (xs, n)) = (\text{cons}(a, xs), \text{wt}(a, \text{head}(xs)) + n)$.

$\text{cost} \triangleq ([\text{wrapz}, \text{consw}]) \pi_2$, $([\text{wrapz}, \text{consw}]) = (1, \text{cost})$, $R \triangleq \text{cost} \leq \text{cost}^\circ$.

$Q \triangleq R \cap (\text{head head}^\circ)$, $S \triangleq F(1, \exists) \alpha$, $\frac{F(\exists, 1)}{\exists} = 1 + \text{cpl}$, $\frac{F(1, \exists)}{\exists} = 1 + \text{cpr}$, $\text{step} \triangleq \text{cpr } P(\text{cons}) \text{ est}(R)$.

(8.2b)

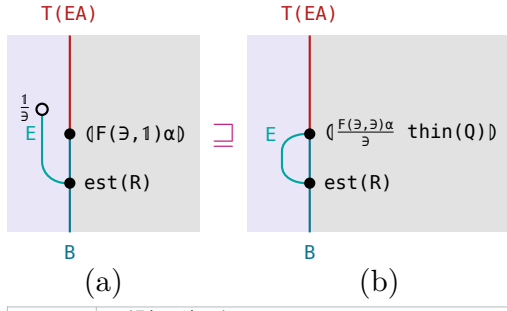
the law	what it says
$F(\exists, Q) \alpha \subseteq F(\exists, 1) \alpha Q$	$F(\exists, 1) \alpha$ is monotonic on Q ; on R it is not, since the next edge can cost arbitrarily much
$S \text{ head} \subseteq [1, \pi_1] \quad S \triangleq F(1, \exists) \alpha$	$S \text{ head}$ is simple, which gives $R \cap (S^\circ S) \subseteq Q$: between two paths S builds from one argument, equal cost and equal head already means Q

$$Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R^\circ \ R^\circ \sqsubseteq R^\circ \wedge \text{MonoAlg}(F(\exists, 1) \ \alpha, Q) \wedge R \sqcap (F(1, \exists) \ \alpha)^\circ \ F(1, \exists) \ \alpha \sqsubseteq Q \implies \quad (8.2c)$$

$$\langle \frac{F(\exists, 1)}{\exists} \ P(\frac{F(1, \exists)}{\exists} \ \alpha \ \text{est}(R)) \rangle \ \text{est}(R) \sqsubseteq \langle \frac{F(\exists, 1)}{\exists} \ \alpha \rangle \ \text{est}(R)$$

a least-cost path in a layered network, as a fold over the layers

$$= \frac{L(\exists)}{\exists} \ \text{est}(R)$$



(a)	$\langle \frac{F(\exists, 1)}{\exists} \ \alpha \rangle \ \text{est}(R)$	Corollary 8.1, (8.2b)
(b)	$\langle \frac{F(\exists, \exists)}{\exists} \ \alpha \ \text{thin}(Q) \rangle \ \text{est}(R)$	

$\sqsubseteq$ (8.2d) under $\langle \rangle$ monotonic: the whole chain runs inside the reduce, and the **est**(R) behind it never moves.

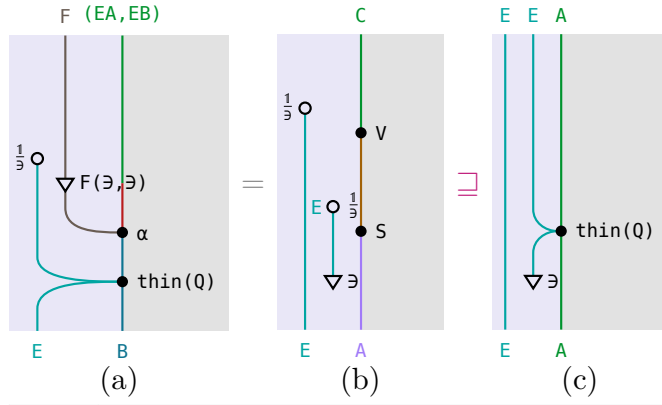
$$R \cap (F(1, \exists) \alpha)^\circ F(1, \exists) \alpha \subseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R)) \subseteq \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q) \quad (8.2d)$$

thinning the algebra of a layered network costs no more than taking the program's two cases

$$\frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists}) \cup P(\text{thin}(Q)) \subseteq \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q)$$

thinning each set the algebra's parts build, then taking the union, is below thinning the one set the whole algebra builds

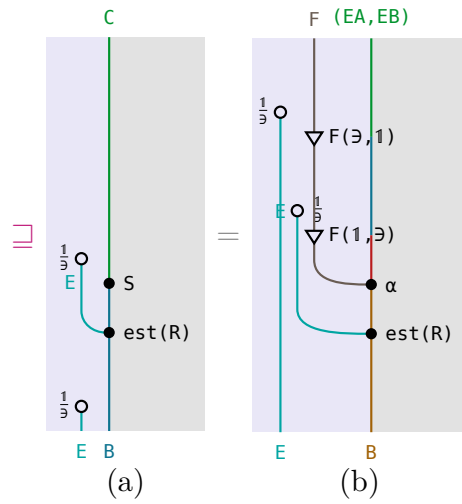
the algebra of (8.2c) row 2



(a)	$\frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q)$	$\frac{F(\exists, 1)F(1, \exists)\alpha = F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) \cup$
(b)	$\frac{V}{\exists} P(\frac{S}{\exists}) \cup$	$\text{union thin}(Q) \supseteq P(\text{thin}(Q)) \cup \text{— (8.1a)}$
(c)	$P(\text{thin}(Q)) \cup$	$\frac{S}{\exists} \text{thin}(Q) \supseteq \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \text{— (8.1a), (8.2b)}$

$$R \cap (F(1, \exists) \alpha)^\circ F(1, \exists) \alpha \subseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R)) \subseteq \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists}) \cup P(\text{thin}(Q)) \cup$$

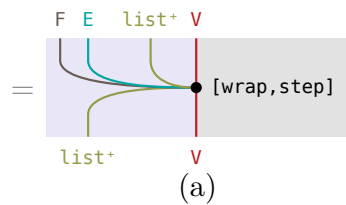
choosing an R-least member of each of those sets is below thinning each and taking the union



(a)	$\frac{S}{\exists} \text{est}(R) \frac{1}{\exists}$	$P(\frac{1}{\exists}) \cup \text{—} 1$
(b)	$\frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R))$	$\alpha \text{ a map; } \frac{F(1, \exists)}{\exists} P(\alpha) \text{ est}(R) = [\text{wrap}, \text{step}] \text{—} \triangle \text{ cost}$

$$\text{cp } P(\alpha) \text{ est}(\text{path}) = [\text{wrap}, \text{step}]$$

at the layered network, choosing an R-least path from each case is the program's two cases $[\text{wrap}, \text{step}]$



(a)	$[\text{wrap}, \text{step}]$	
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$$= \frac{F(\exists, 1)}{\exists} = 1 + \text{cpl} \text{—} \triangle \text{ cost}$$

§8.3. Implementing thin

(8.3a)

$\text{setify} : [A] \rightarrow EA$, $\text{cup} : EA \times EA \rightarrow EA$, $\text{cp}(F) \triangleq \text{subseq}$, $\text{listcp} : F(L) \rightarrow LF$, $\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P$ for P a connected preorder.

$\text{thinlist}(Q)$ is any $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ with $\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q)$; one is $\llbracket \text{nil}, \text{bump } Q \rrbracket$, $\text{bump } Q (a, []) = [a]$, $\text{bump } Q (a, [b] \# xs) = (b \sqsubseteq_Q a \rightarrow [a] \# xs, a \sqsubseteq_Q b \rightarrow [b] \# xs, [a] \# [b] \# xs)$.

Binary thinning data: $S = (f_1 p_1) \cup (f_2 p_2)$ with p_1, p_2 coreflexive; Q a preorder with $Q \sqsubseteq R$ and both $f_1 p_1, f_2 p_2$ monotonic on Q ; P a connected preorder with both f_1, f_2 monotonic on P ; $g_i \triangleq \text{list}(f_i) \text{ filter}(p_i)$.

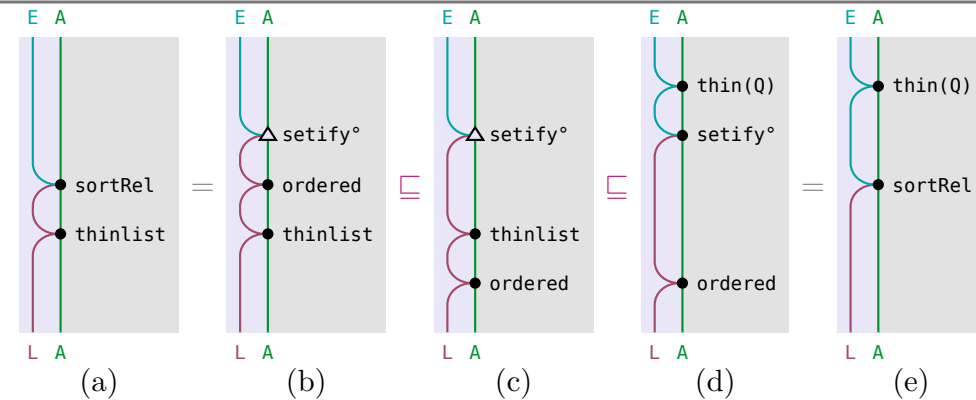
(8.3b)

the law	what it says
$\text{thinlist}(Q) \text{ xs} = [\text{minlist}(Q) \text{ xs}]$ (8.5), Q connected, xs non-empty	what thinning should come to when it can: one element
$\text{sort}(P) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(P)$ (8.6) — (8.6)	thinning a sorted list is a thinning of the set — this is what $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ buys
$\text{sort}(P) \text{ minlist}(Q) \sqsubseteq \text{est}(Q)$ (8.7)	a minimum of the sorted list is a minimum of the set
$\text{sort}(f P f^\circ) \text{ list}(f) \sqsubseteq P(f) \text{ sort}(P)$ (8.8)	shunt a function through a sort
$\text{sort}(P) \text{ filter}(p) \sqsubseteq E(p) \text{ sort}(P)$ (8.9), p coreflexive	filtering a sorted list sorts the restricted set
$(\text{sort}(P) \times \text{sort}(P)) \text{ merge}(P) \sqsubseteq \text{cup sort}(P)$ (8.10)	merging two sorted lists sorts their union
$F(\text{sort}(P)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sort}(FP)$ (8.11), F linear	listcp is the list implementation of the cartesian product $\text{cp}(F)$

Map(setify) $\wedge$ Coreflexive(ordered) $\wedge$ thinlist $\sqsubseteq$ subseq $\wedge$ ordered subseq $\sqsubseteq$ subseq ordered $\wedge$ (8.6)

thinlist setify $\sqsubseteq$ setify thin(Q) $\Rightarrow$ sortRel thinlist $\sqsubseteq$ thin(Q) sortRel

a thinning of the sorted list lists a thinning of the set — P a connected preorder, $\text{thinlist}(Q) \sqsubseteq \text{subseq}$.



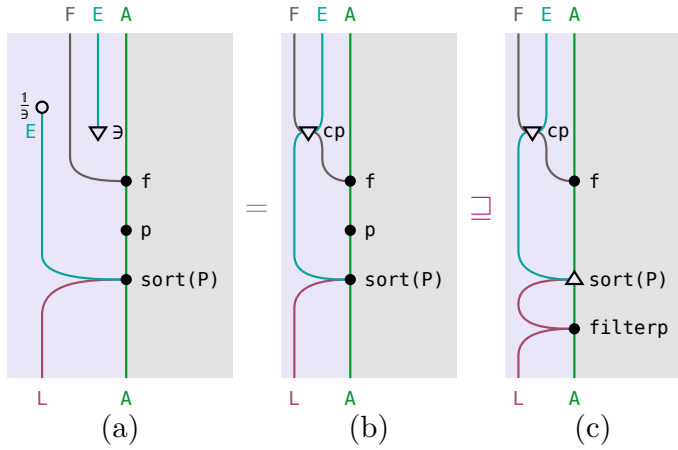
(a)	sortRel thinlist	$\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \text{ — } \triangleq \text{thinlist}$
(b)	$\text{setify}^\circ \text{ ordered thinlist}$	$\text{ordered } P \text{ thinlist}(Q) \sqsubseteq \text{thinlist}(Q) \text{ ordered } P \text{ — } \triangleq \text{thinlist}$
(c)	$\text{setify}^\circ \text{ thinlist ordered}$	$\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q) \text{ — } \triangleq \text{thinlist, dom-laws, triple-chains}$
(d)	$\text{thin}(Q) \text{ setify}^\circ \text{ ordered}$	$\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \text{ — } \triangleq \text{thinlist}$
(e)	$\text{thin}(Q) \text{ sortRel}$	

Map(f) $\wedge$ ($\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)$) $\wedge$ MonoAlg(f,P) $\wedge$ sortF(f P f $^\circ$) listf $\sqsubseteq$ P(f) sort(P) $\wedge$ sort(P) filterp $\sqsubseteq$ E(p) sort(P) $\wedge$ F(sort(P)) listcp $\sqsubseteq$ cp sortF(F(P)) $\Rightarrow$ F(sort(P)) listcp listf filterp $\sqsubseteq$ $\frac{F(\exists) f p}{\exists}$ sort(P) Lemma 8.1

one sorted list built from sorted arguments, instead of a set built and then sorted — **f : FA $\rightarrow$ A** monotonic
on **P**, **p** coreflexive, **F** linear.

Map(f) $\wedge$ sort(P) filterp $\sqsubseteq$ E(p) sort(P) $\Rightarrow$ cp P(f) sort(P) filterp $\sqsubseteq$ $\frac{F(\exists) f p}{\exists}$ sort(P)

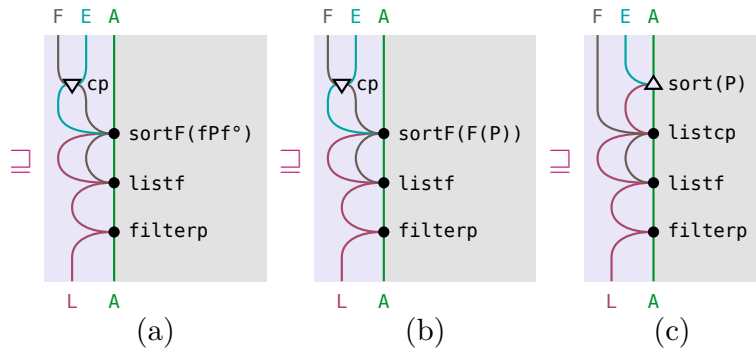
sorting the f-images of the product and filtering by p is below sorting the set F($\exists$)fp builds



(a)	$\frac{F(\exists) f p}{\exists} \text{ sort}(P)$	$\frac{F(\exists) f p}{\exists} = \frac{F(\exists)}{\exists} E(fp)$ — pow-laws, $\triangleq$ thinlist $E(f)=P(f)$, f a map — powrel-laws
(b)	cp P(f) E(p) sort(P)	sort(P) filter(p) $\sqsubseteq$ E(p) sort(P) — (8.3b)
(c)	cp P(f) sort(P) filterp	sort(fPf $^\circ$) list(f) $\sqsubseteq$ P(f) sort(P) — (8.3b)

Map(f) $\wedge$ ($\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)$) $\wedge$ MonoAlg(f,P) $\wedge$ sortF(f P f $^\circ$) listf $\sqsubseteq$ P(f) sort(P) $\wedge$ F(sort(P)) listcp $\sqsubseteq$ cp sortF(F(P)) $\Rightarrow$ F(sort(P)) listcp listf filterp $\sqsubseteq$ cp P(f) sort(P) filterp

combining the arguments' sorted lists by listcp, mapping f and filtering is below sorting the f-images and filtering



(a)	cp sortF(f P f $^\circ$) listf filterp	FP $\sqsubseteq$ fPf $^\circ$ — mon-str, $\triangleq$ thinlist
(b)	cp sortF(F(P)) listf filterp	F(sort(P)) listcp $\sqsubseteq$ cp(F) sort(FP) — (8.3b)
(c)	F(sort(P)) listcp listf filterp	

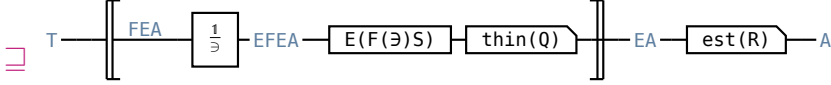
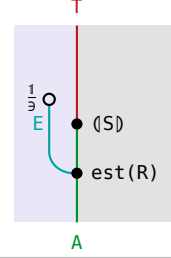
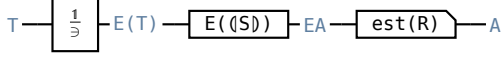
Theorem 8.2

$\text{Map}(f_1) \wedge \text{Map}(f_2) \wedge Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R^\circ \sqsubseteq R^\circ \wedge \text{MonoAlg}(f_1 \text{ } p_1, Q) \wedge$
 $\text{MonoAlg}(f_2 \text{ } p_2, Q) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, P) \wedge$
 $\text{MonoAlg}(f_2, P) \wedge \text{sortF}(f_1 \text{ } P \text{ } f_1^\circ) \text{ listf}_1 \sqsubseteq P(f_1) \text{ sort}(P) \wedge \text{sortF}(f_2 \text{ } P \text{ } f_2^\circ) \text{ listf}_2 \sqsubseteq$
 $P(f_2) \text{ sort}(P) \wedge \text{sort}(P) \text{ filterp}_1 \sqsubseteq E(p_1) \text{ sort}(P) \wedge \text{sort}(P) \text{ filterp}_2 \sqsubseteq E(p_2) \text{ sort}(P)$
 $\wedge F(\text{sort}(P)) \text{ listcp} \sqsubseteq \text{cp sortF}(F(P)) \wedge (\text{sort}(P) \times \text{sort}(P)) \text{ merge}(P) \sqsubseteq \text{cup sort}(P) \wedge$
 $\text{sort}(P) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(P) \wedge \text{sort}(P) \text{ minlist}(R) \sqsubseteq \text{est}(R) \wedge g_1 = \text{listf}_1$
 $\text{filterp}_1 \wedge g_2 = \text{listf}_2 \text{ filterp}_2 \wedge S = f_1 \text{ } p_1 \cup f_2 \text{ } p_2 \Rightarrow (\text{listcp } \langle g_1, g_2 \rangle \text{ merge}(P)$
 $\text{thinlist}(Q)) \text{ minlist}(R) \sqsubseteq \frac{(S)}{\exists} \text{ est}(R)$

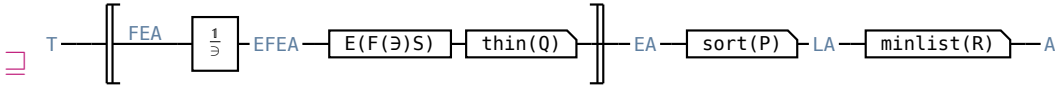
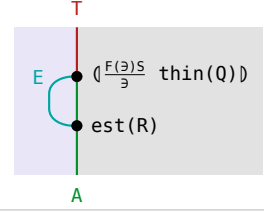
a fold on sorted lists of partial solutions, thinned at every step — at $\triangleq \text{thinlist}$'s binary thinning data.

circuit

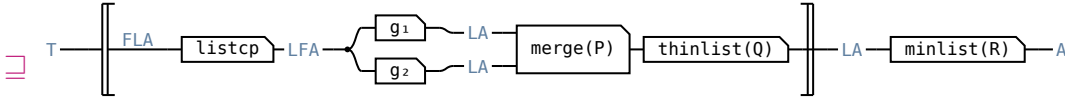
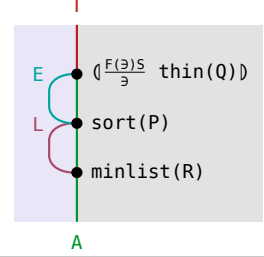
Hinze–Marsden



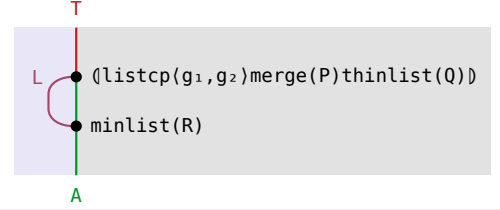
Corollary 8.1 at $f_1 p_1$ and $f_2 p_2$ monotonic on $Q \triangleq \text{thinlist}$



$\text{sort}(P) \text{ minlist}(R) \sqsubseteq \text{est}(R) \text{ — (8.3b) at its } Q \triangleq R$



fusion at (8.3c)



$$\begin{aligned}
& \text{Map}(f_1) \wedge \text{Map}(f_2) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, P) \wedge \text{MonoAlg}(f_2, P) \wedge \quad (8.3c) \\
& \text{sortF}(f_1 P f_1^\circ) \text{ listf}_1 \sqsubseteq P(f_1) \text{ sort}(P) \wedge \text{sortF}(f_2 P f_2^\circ) \text{ listf}_2 \sqsubseteq P(f_2) \text{ sort}(P) \wedge \text{sort}(P) \\
& \text{filterp}_1 \sqsubseteq E(p_1) \text{ sort}(P) \wedge \text{sort}(P) \text{ filterp}_2 \sqsubseteq E(p_2) \text{ sort}(P) \wedge F(\text{sort}(P)) \text{ listcp} \sqsubseteq \text{cp} \\
& \text{sortF}(F(P)) \wedge (\text{sort}(P) \times \text{sort}(P)) \text{ merge}(P) \sqsubseteq \text{cup} \text{ sort}(P) \wedge \text{sort}(P) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \\
& \text{sort}(P) \Rightarrow F(\text{sort}(P)) \text{ listcp} (\text{listf}_1 \text{ filterp}_1, \text{listf}_2 \text{ filterp}_2) \text{ merge}(P) \text{ thinlist}(Q) \sqsubseteq \\
& \quad \frac{F(\exists) (f_1 p_1 \cup f_2 p_2)}{\exists} \text{ thin}(Q) \text{ sort}(P)
\end{aligned}$$

sorting the candidate set is what turns the thinning algebra into an algebra on lists — the side condition of binary thinning theorem's last step.

circuit	Hinze–Marsden
$\sqsubseteq$ $\text{sort}(P) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(P) \quad (8.3b)$	
$=$ $\{ \frac{F(\exists)f_1 p_1}{\exists}, \frac{F(\exists)f_2 p_2}{\exists} \} \text{ cup} \text{ sort}(P) \text{ thinlist}(Q)$ $S = (f_1 p_1) \cup (f_2 p_2) \quad \triangleq \text{thinlist}, \text{ then } \triangleq \text{cup}$	
$\sqsubseteq$ $\{ \frac{F(\exists)f_1 p_1}{\exists}, \frac{F(\exists)f_2 p_2}{\exists} \} (\text{sort}(P) \times \text{sort}(P)) \text{ merge}(P) \text{ thinlist}(Q)$ $(\text{sort}(P) \times \text{sort}(P)) \text{ merge}(P) \sqsubseteq \text{cup} \text{ sort}(P) \quad (8.3b)$	
$=$ $\{ \frac{F(\exists)f_1 p_1}{\exists} \text{ sort}(P), \frac{F(\exists)f_2 p_2}{\exists} \text{ sort}(P) \} \text{ merge}(P) \text{ thinlist}(Q)$ $\langle X, Y \rangle (\text{sort}(P) \times \text{sort}(P)) = \langle X \text{ sort}(P), Y \text{ sort}(P) \rangle \quad \text{— bdm-prod-laws}$	
$\sqsubseteq$ $F(\text{sort}(P)) \text{ listcp} \langle g_1, g_2 \rangle \text{ merge}(P) \text{ thinlist}(Q)$ Lemma 8.1 at f_1, p_1 and at f_2, p_2, then $X \langle g_1, g_2 \rangle \sqsubseteq \langle Xg_1, Xg_2 \rangle \quad \text{— bdm-prod-laws; } g_i \triangleq \text{list}(f_i)$ $\text{filter}(p_i) \quad \triangleq \text{thinlist}$	

§8.4. The knapsack problem

(8.4a)

$\text{vol}, \text{wt} : \text{Item} \rightarrow \text{Real}, \text{ value} \triangleq \text{list}(\text{vol}) \text{ sum}, \text{ weight} \triangleq \text{list}(\text{wt}) \text{ sum}.$

$\text{subseq} = ([\text{nil}, \text{cons}] \cup [\text{nil}, \pi_2])^\circ$, within w the coreflexive on xs with weight $xs \leq w, 0 \leq w$.

$R \triangleq \text{value} \geq \text{value}^\circ, Q \triangleq R \cap (\text{weight} \leq \text{weight}^\circ), P \triangleq R, .$

$\text{FA} = 1 + \text{Item} \times A, \quad \text{listcp} = \text{wrap} + \text{cpr}, \quad g_1 \triangleq \text{list}([\text{nil}, \text{cons}]) \quad \text{filter}(\text{within } w) \quad = [\text{list}(\text{nil}), h_1],$
 $g_2 \triangleq \text{list}([\text{nil}, \pi_2]) = [\text{list}(\text{nil}), h_2].$

$h_1 \triangleq \text{list}(\text{cons}) \text{ filter}(\text{within } w), \quad h_2 \triangleq \text{list}(\pi_2).$

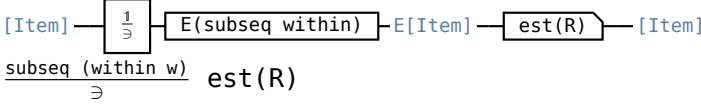
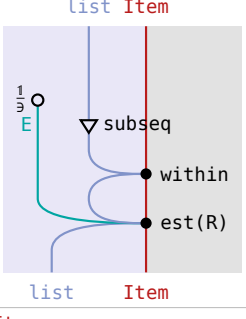
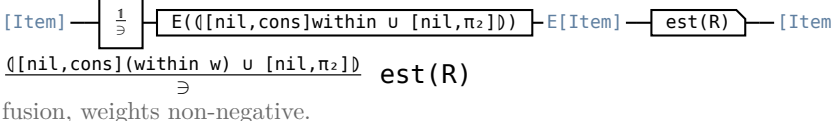
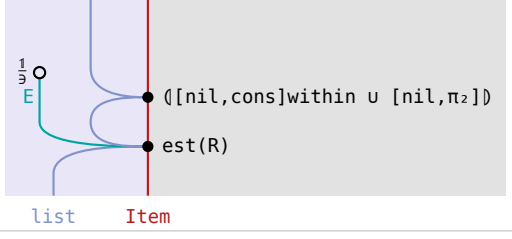
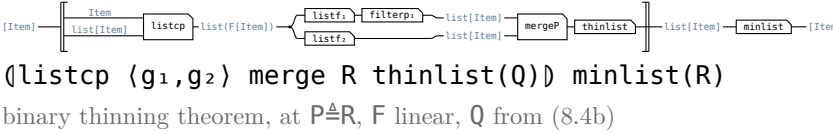
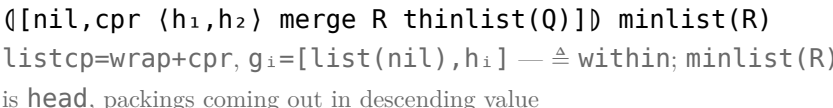
(8.4b)

the law	what it says
$(1 \times R) (\text{cons } (\text{within } w)) \sqsubseteq \text{cons } (\text{within } w) R$ FALSE	not monotonic on R: a selection of greater value need not still fit once one more item goes in
$(1 \times Q) (\text{cons } (\text{within } w)) \sqsubseteq \text{cons } (\text{within } w) Q$ $(1 \times Q) \pi_2 \sqsubseteq \pi_2 Q$,	both halves are monotonic on Q once ties in value are broken by weight

(8.4c)

$$\begin{aligned}
& 0 \leq w \wedge (\forall i. 0 \leq \text{wt}(i)) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{nil}, \text{cons}] R \\
& [\text{nil}, \text{cons}]^\circ) \text{listf}_1 \sqsubseteq P([\text{nil}, \text{cons}]) \text{sortP} \wedge \text{sortF}([\text{nil}, \pi_2] R [\text{nil}, \pi_2]^\circ) \text{listf}_2 \sqsubseteq P([\text{nil}, \pi_2]) \\
& \text{sortP} \wedge \text{sortP} \text{filterp}_1 \sqsubseteq E(\text{within}) \text{sortP} \wedge F(\text{sortP}) \text{listcp} \sqsubseteq \text{cp sortF}(F(R)) \wedge (\text{sortP} \times \text{sortP}) \\
& \text{mergeP} \sqsubseteq \text{cup sortP} \wedge \text{sortP} \text{thinlist} \sqsubseteq \text{thin}(Q(\text{vol}, \text{wt})) \text{sortP} \wedge \text{sortP} \text{minlist} \sqsubseteq \text{est}(R) \Rightarrow \\
& (\text{listcp } (\text{listf}_1 \text{filterp}_1, \text{listf}_2) \text{mergeP thinlist}) \text{minlist} \sqsubseteq \frac{\text{subseq within}}{\supset} \text{est}(R)
\end{aligned}$$

the knapsack problem, as a fold that thins the packings kept at each item

circuit	Hinze–Marsden
 $\frac{\text{subseq } (\text{within } w)}{\supset} \text{est}(R)$	
 $\frac{([\text{nil}, \text{cons}] (\text{within } w) \cup [\text{nil}, \pi_2])}{\supset} \text{est}(R)$ fusion, weights non-negative.	
 $(\text{listcp } (g_1, g_2) \text{merge } R \text{thinlist}(Q)) \text{minlist}(R)$ binary thinning theorem, at $P^A R$, F linear, Q from (8.4b)	
 $([\text{nil}, \text{cpr } (h_1, h_2) \text{merge } R \text{thinlist}(Q)]) \text{minlist}(R)$ $\text{listcp} = \text{wrap} + \text{cpr}, g_i = [\text{list}(\text{nil}), h_i] \text{ --- } \triangleq \text{within}; \text{minlist}(R)$ is head, packings coming out in descending value	

§8.5. The paragraph problem

(8.5a)

```

Line=list+ Word, Para=list+ Line, FA=Word+Word×A, listcp=wrap+cpr.
new(a,xs)=[a]#xs, glue(a,xs)=[a]#head(xs)#tail(xs), partition≐([wrap wrap,new u glue]) : list+
Word→Para.

width≐([length,(length×1) plus succ]), 0≤length a, fits w the coreflexive on a line x with width x≤w, ok w the
coreflexive on [x]#xs with width x≤w.

white w x=w-width x, collect≐list(sqr) sum, waste w≐init list(white w) collect.

R≐(waste w)≤(waste w)°, Q≐Rn(head head°), P≐T, .

g1≐list([wrap wrap,new]), g2≐list([wrap wrap,glue]) filter(ok w), start≐wrap wrap wrap, h1≐list(new),
h2≐list(glue) filter(ok w).

```

(8.5b)

the law	what it says
$(1 \times R) \text{ glue} \sqsubseteq \text{glue } R \quad \text{FALSE}$	glue is not monotonic on R: the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this
$(1 \times Q) \text{ new} \sqsubseteq \text{new } Q$ $(1 \times Q) (\text{glue } (ok \ w)) \sqsubseteq \text{glue } (ok \ w) Q$ $\text{cons monotonic on } \text{collect} \sqsubseteq \text{collect}^\circ,$	both halves are monotonic on Q once ties in waste are broken by the first line
merge $T = \text{cat}$; $P \triangleq \text{head prefix head}^\circ$ also serves	T needs no sorting at all, and prefix is a linear order on first lines of paragraphs of one input

(8.5c)

$$\begin{aligned}
& (\forall a. 0 \leq \text{len}(a)) \wedge (\forall a. \text{len}(a) \leq w) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{wrap} \\
& \text{wrap,new}] \sqsubseteq [\text{wrap wrap,new}]^\circ) \text{listf}_1 \sqsubseteq P([\text{wrap wrap,new}]) \text{sortP} \wedge \text{sortF}([\text{wrap wrap,glue}] \sqsubseteq \\
& [\text{wrap wrap,glue}]^\circ) \text{listf}_2 \sqsubseteq P([\text{wrap wrap,glue}]) \text{sortP} \wedge \text{sortP filterp}_2 \sqsubseteq E(ok) \text{sortP} \wedge \\
& F(\text{sortP}) \text{listcp} \sqsubseteq \text{cp sortF}(F(T)) \wedge (\text{sortP} \times \text{sortP}) \text{mergeP} \sqsubseteq \text{cup sortP} \wedge \text{sortP thinlist} \sqsubseteq \\
& \text{thin}(Q(\text{len},w)) \text{sortP} \wedge \text{sortP minlist} \sqsubseteq \text{est}(R) \Rightarrow ([\text{listcp } \langle \text{listf}_1, \text{listf}_2 \text{ filterp}_2 \rangle \text{ mergeP} \\
& \text{thinlist}) \text{minlist} \sqsubseteq \frac{\text{partition fits}}{\exists} \text{est}(R)
\end{aligned}$$

a paragraph laid out as a fold that thins the layouts kept at each word

circuit	Hinze–Marsden
$ \begin{aligned} & \text{list}^*(\text{Word}) \xrightarrow{\frac{1}{2}} E(\text{list}^*(\text{Word})) \xrightarrow{E([\text{wrap wrap,new} u [\text{wrap wrap,glue}ok])} E(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word})) \\ & \frac{\text{partition list}^*(\text{fits } w)}{\exists} \text{est}(R) \end{aligned} $	
$ \begin{aligned} & \text{list}^*(\text{Word}) \xrightarrow{\frac{1}{2}} E(\text{list}^*(\text{Word})) \xrightarrow{E([\text{wrap wrap,new} u [\text{wrap wrap,glue}ok])} E(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word})) \\ & = \frac{([\text{wrap wrap,new} u (\text{glue } (ok \ w))])}{\exists} \text{est}(R) \\ & \text{fusion, every word fits on a line by itself.} \end{aligned} $	
$ \begin{aligned} & \text{list}^*(\text{Word}) \xrightarrow{\frac{1}{2}} E(\text{list}^*(\text{Word})) \xrightarrow{E([\text{wrap wrap,new} u [\text{wrap wrap,glue}ok])} E(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word})) \\ & = \frac{([\text{wrap wrap,new} u ([\text{wrap wrap,glue} (ok \ w))])}{\exists} \text{est}(R) \\ & \text{the algebra as } (f_1 p_1) \cup (f_2 p_2), p_1 \triangleq 1 \text{ — } \triangleq \text{thinlist.} \end{aligned} $	
$ \begin{aligned} & ([\text{listcp } \langle g_1, g_2 \rangle \text{ cat thinlist}(Q)] \text{minlist}(R) \\ & \text{binary thinning theorem, at } P \triangleq T \text{ with merge } T = \text{cat}, Q \text{ from} \\ & (8.5b) \end{aligned} $	
$ \begin{aligned} & ([\text{start}, \text{cpr } \langle h_1, h_2 \rangle \text{ cat thinlist}(Q)] \\ & = \text{minlist}(R) \\ & \text{listcp} = \text{wrap} + \text{cpr}, g_i \text{ along the coproduct — } \triangleq \text{partition} \end{aligned} $	

FA=(City×City)+(City×A), the base functor of cons-lists of length at least two; listcp=wrap+cpr; tc : City×City→Real, neither positive nor symmetric.

$$\text{cost}(x_s, y_s) = \text{outcost } x_s + \text{incost } y_s, \quad \text{outcost}[a_0, \dots, a_n] = \text{tc}(a_0, a_1) + \dots + \text{tc}(a_{n-1}, a_n), \quad \text{incost}[a_0, \dots, a_n] = \text{tc}(a_1, a_0) + \dots + \text{tc}(a_n, a_{n-1}).$$

```
next $\hat{=}$ tail head, next2 $\hat{=}$ next $\times$ next, head2 $\hat{=}$ head $\times$ head, R $\hat{=}$ cost $\leq$ cost $^\circ$ , Q $\hat{=}$ Rn(next2 next2 $^\circ$ )n(head2 head2 $^\circ$ ), P $\hat{=}$ T,
g1 $\hat{=}$ list([start,dropl]), g2 $\hat{=}$ list([start,dropr]).
```

the law	what it says
$(1 \times R) \text{ drop}l \sqsubseteq \text{drop}l \text{ } R$ FALSE $(1 \times R) \text{ drop}r \sqsubseteq \text{drop}r \text{ } R$ FALSE	neither drop is monotonic on R : the two edges it adds and removes depend on head and next of both lists
$(1 \times Q) \text{ drop}l \sqsubseteq \text{drop}l \text{ } Q$ $(1 \times Q) \text{ drop}r \sqsubseteq \text{drop}r \text{ } Q$,	both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input

a least-cost bitonic tour, as a fold that thins the tours kept at each city

Hinze–Marsden

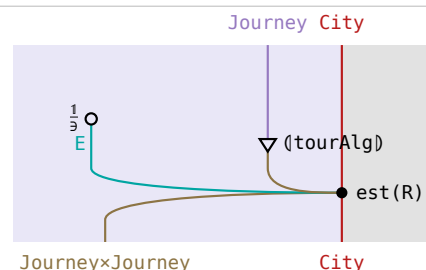


Diagram illustrating the relationship between Journey and City. A vertical line separates the Journey space (left, light blue) from the City space (right, light grey). A teal line labeled "list" starts in Journey and ends at a point in City labeled $(\text{listcp}(g_1, g_2) \text{cat thinlist})$. A blue line starts in Journey and ends at a point in City labeled "minlist". The bottom of the diagram is labeled "Journey $\times$ Journey" on the left and "City" on the right.

$$dL(City:City, City) \left[\begin{array}{c} \boxed{F(list(Journey^*))} \\ \boxed{listcp} \end{array} \begin{array}{c} \text{list}(F(Journey^*)) \\ \text{list}(F(Journey^*)) \end{array} \left[\begin{array}{c} \boxed{g_1} \\ \boxed{g_2} \end{array} \begin{array}{c} \text{list}(Journey^*) \\ \text{list}(Journey^*) \end{array} \right] \boxed{cat} \boxed{thinlist} \begin{array}{c} \text{list}(Journey^*) \\ \text{list}(Journey^*) \end{array} \right] \boxed{minlist} \begin{array}{c} \text{Journey} \\ \text{Journey} \end{array}$$

$$(listcp \ (g_1, g_2) \ cat \ thinlist(Q)) \ minlist(R)$$

binary thinning theorem, at $P^{\Delta}T$ with $merge \ T=cat, Q$ from (8.6b)

```

([start wrap,cpr (list(dropl),list(dropr)) cat
thinlist(Q)]) minlist(R)
listcp=wrap+cpr, gi=[list(start),list(dropi)] —  $\triangleq$  tour;
quadratic, two tours added per step

```